# Solutions to Introductory Quiz in Probability

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**Abstract**

Solutions to last time's introductory quiz with brief comments for each one. Also, do not miss the minor challenges in some of the most (or least) interesting questions!

## Question 1

> We toss a fair coin three times. The probability of HHH (three times "heads") is less than the probability of HTH ("heads", followed by "tails", followed by "head").

**True.** This is a relatively simple question on classical probability. Simple counting works miracles here, since there is just one event where all tosses yield H, and there is also exactly one series of tossses resulting to HTH, so the two probabilities must be equal (given the fairness of the used coin).

In particular, the corresponding probability is: $\frac{1}{8}$.

## Question 2

> We toss a fair coin three times. The probability of seeing H twice is greater than the probability of seeing H trice.

**True.** Again, this boils down to simple counting. There is only one event where we have HHH (HHH itself), while there are three events where we encounter exactly two heads, namely: HHT, HTH, THH. So, since $3 > 1$, there are more chances to see H twice than trice.

*Twist:* What if we tossed the coin 5 times? Is seeing H twice or trice the most possible outcome?

## Question 3

> We roll two fair dices. What is the probability of both dices bringing the same result?

The correct answer is $\frac{1}{6}$ as most of you have correctly responded. The total number of combinations is $6 \cdot 6 = 36$, while those of interest are exactly $6$ (both dices facing up the same value). So, in total, we have:

$$\frac{6}{36} = \frac{1}{6}.$$

## Question 4

> In a multiple choice test each question has 4 possible answer, with exactly one of those being correct. A particular participant knows the correct answer in a question with probability 0.70. Then, the probability of knowing the answer having answered correctly is...

The correct answer is $0.90$. The easiest way to see this is by using conditional probabilities, so, let $A$ be the event that the can answer correctly and $B$ the event that the student answers correctly. In that setting, we already know that $p(A) = 0.70$ and we are interested in $p(A \mid B)$. From the Total Probability Theorem (which is no more than just partitioning the event space) we get:

$$p(B) = p(B \cap A) + p(B \cap A') = p(B \mid A)p(A) + p(B \mid A')p(A').$$

So, the probability of the student answering correctly in a question they do know ($B \cap A$) is $0.70$, as we already know, while the probability they answer correctly a question they do not know is $\frac{1}{4} \times 0.30 = 0.075$. So, in total:

$$p(B) = 0.70 + 0.075 = 0.775.$$

Then, we have:

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)} = \frac{0.70}{0.775} \approx 0.90.$$

### Another line of attack...

We can also answer the above by counting. The percentage of questions answered correctly will be $70\%$ plus one fourth of the other $30\%$ the student cannot answer correctly, so they answer at random. That is, the percentage of the correctly answered questions is roughly $77.5\%$, out of which the student knows the answer in $70/77.5$ questions, which is the same ratio as the one we obtained above. So, even if you were not aware of conditional probabilities and Bayes' formula, you could work things out using just... counting!

## Question 5

> We toss a fair coin 4 times. What is the probability of getting 3 times H and one time T?

The correct answer is $\frac{1}{4}$. Indeed, in order to get 3 H and 1 T we have four possible combinations, depending on the position at which T appears (1st, 2nd, 3rd or 4th). So, 4 out of the $2^4 = 16$ possible outcomes in total, i.e.:

$$\frac{4}{16} = \frac{1}{4}.$$

## Question 6

> We toss a "soft" coin twice. The first time, the coin behaves as a typical fair coin, while the second, if the result is H, then the probability of getting again H is 0.70, but if it is T, then the probability of getting again H is 0.40. What is the probability of getting one H and one T in those two tosses?

The correct answer is $0.35$. Indeed, we have to discriminate between the two results the first toss yields. So, let $H_i, T_i$ be the events of Heads / Tails in the $i$-th toss, respectively, for $i = 1, 2$ and $H, T$ being the events of getting one time Heads / Tails repsectively. So, we are interested in the probability:

$$p(HT) = p(H_1T_2) + p(H_2T_1).$$

We can compute this in each case by taking into account the coin's softness, as follows:

- In the case the first toss is H, which happens with probability $p(H_1) = 0.50$, we then get T with probaiblity $p(T_2) = 0.30$, so we get $p(H_1T_2) = 0.50 \cdot 0.30 = 0.15$.
- Similarly, in case the first toss is T, with probability $p(T_1) = 0.50$, then we get $H$ with probability $p(H_2) = 0.40$, so $p(T_1H_2) = 0.50 \cdot 0.40 = 0.20$.

In total, we obtain $p(HT) = 0.15 + 0.20 = 0.35$ as promised.

## Question 7

> The intersection of two sets A, B, corresponds to the set of all elements belonging to both A and B.

**True**, by the very definition of the intersection, $A \cap B$ of two sets.

## Question 8

> The union of two sets A, B, corresponds to the set of all elements belonging either to A or B but not both.

**True**, again by the definition of $A \cup B$. Observe that in set theory, the union always includes the intersection of the same two sets, i.e.:

$$A \cap B \subseteq A \cup B$$

which might contradict the common everyday use of "or" as "either... or...". Bear this in mind as you formulate problems from natural language to mathematics.

## Question 9

> The average height of a football team is 1.76. Selling a player of height 1.83 and buying a player of height 1.81 will...

**lower the team's average height.** This is expected, since we remove a larger observation from our sample than the one we introduce. In particular, taking some time to do the math, let $S$ denote the sum of heights of the $n$ players of the team. Then, the new total height will be given by:

$$S' = S - 1.83 + 1.82 = S - 0.02,$$

so the new average value will be:

$$\mu' = \frac{S'}{n} = \frac{S - 0.02}{n} = \frac{S}{n} - \frac{0.02}{n} < \frac{S}{n} = \mu.$$

So, indeed, $\mu' < \mu$.

## Question 10

> Let X be a sample of 37 observations with mean value 23. Multiplying all sample observations by -3, the new average will be...

**-69.** It is a given fact that the mean value is sensitive to linear transformations, i.e., that the mean value of $aX + b$ equals $a\mu + b$ where $\mu$ is the mean value of $X$. Hence, in our case:

$$\mu' = -3\mu = -3 \cdot 23 = -69.$$

## Question 11

> Students of a Probability course score on average 13 points in a 0-20 scale, with a standard deviation of 5.84. Assuming we convert all grades to a 0-100 scale, the standard corresponding mean value, $\mu$, and standard deviation, s, will be...

$\mu = 65, s = 29.20$. As we discussed before, the mean value is sensitive to linear transformations, while standard deviation is sensitive only to scaling, i.e., if $s$ is the standard deviation of a random variable $X$, then the standard deviaation of $aX + b$ will be:

$$s' = |a|s.$$

*As a piece of intuition, think of the sample as a piece of dough. In this analogy, mean value corresponds to its centre of mass and standard deviation to the distribution of the dough around its centre of mass. Thus, moving the dough around naturally moves its centre of mass (mean value) but does not affect its shape (standard deviation), while if we scale the dough, then naturally its centred of mass is appropriately scaled, e.g., when the dough puffs, while this is also true for its shape, which is naturally affected by this transformation.*

## Question 12

> In a sample of 183 observations, we add a new value equal to the maximum observation in the sample. Which of the following statistical functions is not guaranteed to increase?

The **median** is, in general, the least vulnerable to extreme observations since it captures the point in the sample which evenly splits the sample. All other measures mentioned, given that they involve, among others, both extreme observations, are naturally affected by new extreme observations. For a demonstration, consider the following sample of 7 observations:

$$1, 1, 2, 4, 4, 7, 8.$$

Then, adding another extreme value, e.g., $2027$, will result to the range, mean value and standard deviation increasing significantly (you can verify this computationally), but the median remains intact at $\delta = 4$ in both samples.

## Question 13

> The mean value of sample is not sensitive to extreme observations.

**False**, as illustrated by the previous example (see Question 12).

## Question 14

> We choose two balls from a bowl containing 12 black and 8 white ones without replacement. The probability of getting two balls of the same colour is...

**192/380.** To see this, let us denote by $B_i, W_i$ the events of drawing a black and white ball at iteration $i = 1, 2$, respectively. Then, we are interested in the probability:

$$p(B_1W_2 \cup W_1B_2) = p(B_1W_2) + p(W_1B_2).$$

Given that there is no replacement, i.e., each ball extracted is then removed from the bowl, we easily get:

$$p(B_1W_2) = \frac{12}{20}\frac{8}{19}.$$

Similarly:

$$p(W_1B_2) = \frac{8}{20}\frac{12}{19}.$$

So, in total ($BW$ denotes the event of interest, just for simplicity):

$$p(BW) = \frac{12}{20}\frac{8}{19} + \frac{8}{20}\frac{12}{19} = \frac{192}{380}.$$

*You might observe some symmetry between the two cases examined above. Is this a generalisable observation or is it coincidental?*

## Question 15

> We choose two balls from a bowl containing 12 black and 8 white ones with replacement. The probability of getting two balls of the same colour is...

**192/400.** The idea is identical to the previous question, just taking into account that now that we replace balls into the bowl, all denominators are equal to 20.

## Question 16

> We choose two numbers (not necessarily different}, x, y, uniformly at random from the set {1,2,. . . ,100}. Then, their sum, x + y, is uniformly distributed on the set {2,3,. . . ,200}.

**False.** This is a classic in statistics / probability related interviews, so. . . heads up! Even if $x, y$ are drawn from a uniform distribution, i.e., completely at random, the distribution of their sum is not uniform. To see this, observe that, e.g., $x + y = 2$ can be obtained only through $x = y = 1$, while for $x + y = 100$ we have the following combinations:

$$(x, y) \in \{(1, 99), (2, 98), \ldots, (99, 1)\}.$$

In general, the distribution of $x + y$ is what we refer to as *triangular* distribution, since the middle values can be obtained with higher probability for combinatorial reasons.

## Question 17

> A flu test is positive with probability 0.95 given that the patient is actually sick, and with probability 0.20 if the patient is not actually sick. In a population where about 3% of the population is sick, what is the probability that a person tests positive?